

Correcting a learned invariant improves rollouts in a pixel-trained world model

Abstract

World models can predict what happens next without necessarily learning the physics behind it. We ask a stronger question than whether physical quantities can be decoded from a model’s latent state: does the model learn something its own dynamics actually preserve? In DreamerV3 trained only on pendulum video, we find a reproducible energy-like invariant without using physical labels, while the same search fails on matched damped systems where no such invariant should exist. The invariant begins to drift when the model imagines forward on its own, and correcting that drift improves its predictions; random corrections usually make them worse. This suggests that world models can learn real physical structure but still fail to respect it during rollout, and that preserving learned invariants may be a useful way to make their predictions more reliable.

1 Introduction

World models learn to predict future observations and can use those predictions for planning. Good video prediction, however, does not by itself show that a model has learned the underlying dynamics of the system it observes. That distinction matters most when we ask the model to predict far beyond the trajectories it saw during training.

A common way to look for physical structure is to fit a probe from the hidden state to a physical variable. Probe accuracy tells us what information a latent representation retains, but not whether the model’s transition uses that information. Our own baseline makes this problem concrete. Six randomly initialized DreamerV3 models contain polynomial functions of their latent state that correlate with pendulum energy at up to 0.908. A strong correlation can therefore appear even before learning.

We ask a stronger question: does a trained model contain a scalar function of its latent state that its own transition approximately preserves, and does preserving that scalar improve prediction? We study DreamerV3 [Hafner et al., 2023] trained on 64×64 video from a Gymnasium pendulum, with no actor, critic, or physical supervision. After training, we freeze the model and analyze its latent dynamics.

We test the claim in three ways.

1. **Recovery.** At an extraction dimension fixed before evaluating these models, three independently trained conservative models recover scalars with energy correlations of 0.973, 0.967, and 0.975. The recovered scalar has within-trajectory variance equal to 1.1×10^{-4} of its total variance along observation-conditioned trajectories. Six untrained models span 0.170–0.908. Training therefore makes the recovered structure reproducible; it does not simply make energy decodable.
2. **Dissipative control.** We apply the same search to models trained on a damped pendulum, with the damping level fixed by a rule chosen before training. Within the same degree-4 polynomial family, the search finds no approximately conserved candidate. Four statistics separate the conservative and damped models without seed overlap.
3. **Intervention.** The recovered scalar drifts when the model imagines forward without new observations. Projecting the latent state back onto the initial level set of that scalar lowers rollout error by 2.9–3.5% on all three conservative models. Twenty matched random polynomial constraints usually increase error. The recovered constraint beats every random draw on two models; on the third, the specificity result is weak.

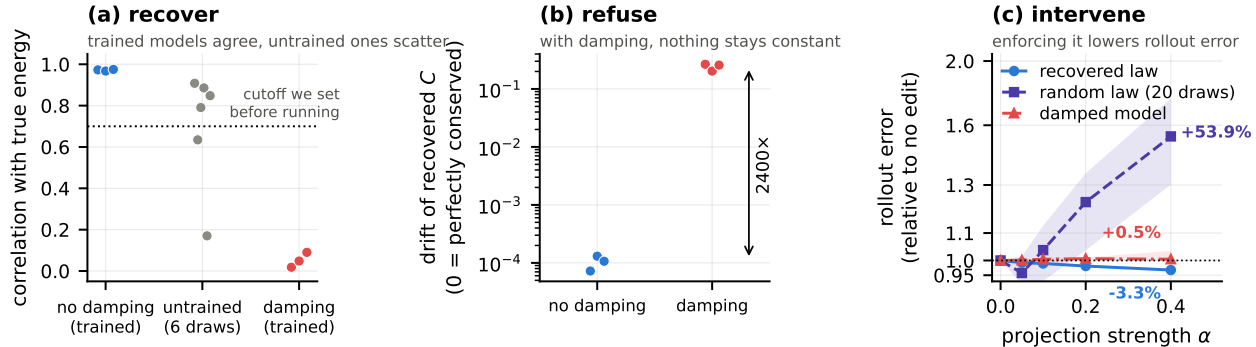


Figure 1: Each point represents one model. **(a)** The three trained conservative models recover energy correlations between 0.967 and 0.975, while six untrained models span 0.17–0.91. **(b)** The recovered scalar changes very little within observation-conditioned trajectories for conservative models and much more for damped models. **(c)** Rollout error as a function of projection strength α , relative to no edit. Enforcing the recovered constraint lowers error; enforcing a matched random polynomial constraint usually raises it; applying the recovered constraint to a damped model changes little.

Two additional results narrow the interpretation. Basis-independent properties of the recovered scalar reproduce across seeds more reliably than selectors tied to individual latent coordinates. The flow-generation term in our extraction rule changes average recovery little but reduces seed-to-seed variation. We also find that accumulated invariant error predicts rollout degradation, but our data do not distinguish a frequency-weighted version of that mechanism from a simpler unweighted one.

2 Setup and method

Model and data. We analyze the published DreamerV3 architecture [Hafner et al., 2023]: 13.5M parameters, categorical 32×32 stochastic latents, a convolutional encoder and decoder, Kullback–Leibler balancing, and unimix. We train only the world model, with no actor or critic.

The data come from Gymnasium’s **Pendulum-v1** [Towers et al., 2024]. Each simulator step renders a 500×500 frame. We crop it to 448×448 and block-average to 64×64 . We set the simulator state directly, sampling $\theta \sim U[-1.8, 1.8]$ and $\dot{\theta} \sim U[-2.2, 2.2]$. We reject any trajectory that reaches the simulator’s $|\dot{\theta}| = 8$ speed clip because clipping would break conservation. Actions are always zero. A single frame does not reveal velocity, so the model must infer $\dot{\theta}$ from the sequence.

We train on a fixed dataset of 256 trajectories, each 120 frames long, using Adam at 10^{-4} , batch size 16, and sequence length 64. We cap each run at 30 minutes of wall-clock time, which gives about 6,500 gradient steps. Three independently trained seeds all pass checks fixed before analysis: KL divergence above 1 nat, one-step decoding at least $4\times$ better than predicting the dataset mean, and finite rollouts. Appendix B gives the remaining training details. Our claims concern a DreamerV3 world model trained on pendulum video, not DreamerV3 as a full reinforcement-learning agent.

Three kinds of latent trajectories. We use three different latent trajectories and keep them separate throughout the paper.

An *observation-conditioned* trajectory is the sequence of deterministic recurrent states produced when the model receives the next real frame at every step. This is the latent sequence returned by the encoder on held-out video.

The *one-step transition* T is the model’s autonomous recurrent update with zero action and no new observation. It maps the current recurrent state to the next one.

An *imagination rollout* starts from one encoded state and repeatedly applies T without supplying any later frames.

To estimate the latent flow, we apply T once at each state from an observation-conditioned trajectory. The transition is therefore autonomous even though the states at which we evaluate it come from real video.

Sections 3 and 4 measure conservation along observation-conditioned trajectories. Section 5 then tests what happens when the same scalar drifts during autonomous imagination.

Latent coordinates. After training, we freeze the model. Let h denote the deterministic recurrent state on held-out video and $\bar{h}$ its mean. We take the top 12 principal directions of h , remove any direction with no support in the data, and let P denote the resulting orthonormal map. All extraction, projection, and perturbation operate on

$$z = P^\top(h - \bar{h}). \quad (1)$$

Let $F(z)$ be the projection through P of the model’s one-step displacement $T(h) - h$. We use the model’s own transition displacement directly rather than fitting a separate vector field and analyzing that surrogate.

The candidate family. We look for the invariant among degree-4 polynomials in z . Writing $\varphi(z)$ for the vector of monomials up to degree 4 in the 12 latent coordinates, a candidate is $C(z) = a^\top \varphi(z)$, and the search is over the coefficient vector a . There are 1819 such monomials, from z_1 through $z_1 z_2 z_3 z_4$; we exclude the constant term, which is conserved trivially. Degree 4 is a compromise. The pendulum’s energy is quadratic in $\dot{\theta}$ and needs polynomial terms to approximate $\cos \theta$, and the latent is some unknown nonlinear encoding of $(\theta, \dot{\theta})$ rather than those coordinates themselves, so the family has to be rich enough to express energy through that distortion while staying small enough to fit.

The invariance objective. An invariant is constant along any one trajectory and differs between trajectories. We therefore score a candidate by

$$\text{ratio}(C) = \frac{\text{mean within-trajectory variance of } C}{\text{total variance of } C}, \quad (2)$$

which is 0 for a perfectly conserved scalar and near 1 for one that wanders as much within a trajectory as across the dataset. The denominator is what makes the objective non-trivial: without it, $C = 0$ would score perfectly while distinguishing nothing.

Both variances are quadratic forms in a . With W the mean within-trajectory covariance of the features and T their total covariance, Equation 2 is $a^\top W a / a^\top T a$, a Rayleigh quotient. Its minimiser is therefore not something we search for iteratively: it is the eigenvector of the generalized problem $W a = \lambda T a$ with the smallest eigenvalue, and λ is the invariance ratio itself. One eigendecomposition returns the whole family, ranked from most to least conserved.

Requiring the scalar to generate the flow. Conservation alone does not pick out a unique scalar. If C is conserved then so is any function of it, so E , E^2 and mixtures of E with other conserved quantities all sit near the top of the ranking, and the leading eigenvector is generally some combination of them rather than energy. We therefore ask C to do more than stay constant: we ask it to generate the motion. We fit C jointly with an antisymmetric operator B such that

$$F \approx B \nabla C, \quad B^\top = -B. \quad (3)$$

This is Hamiltonian mechanics written without coordinates. In $\dot{q} = \partial H / \partial p$, $\dot{p} = -\partial H / \partial q$, the state moves orthogonally to the energy gradient, and multiplication by an antisymmetric matrix is what rotating a gradient that way looks like in matrix form. The relevant identity is that $\nabla C^\top B \nabla C = 0$ for any antisymmetric B , so if $F = B \nabla C$ then C is constant along the flow by construction rather than by fitting. The condition also separates candidates that the invariance ratio cannot: E^2 is exactly as conserved as E but does not pair with the same B , so the requirement selects a direction inside the conserved family [Arnold, 1989].

The fit is bilinear, so fixing either a or B makes the other a least-squares problem and we alternate. It searches within the top eight eigenvectors from Equation 2, so the recovered C has eight effective degrees of freedom rather than 1819. That matters for a search run on 52 trajectories: a free fit over the full basis could drive the in-sample ratio to zero by overfitting. Section 7 tests what the second condition contributes, and the answer is less than its prominence here suggests.

Table 1: The same extraction pipeline applied to conservative and damped models. Each range covers three seeds. No conservative seed overlaps any damped seed on any row.

statistic	conservative	damped	median ratio
$ \rho _E$	0.967–0.975	0.018–0.090	20×
drift of C	$0.7\text{--}1.3 \times 10^{-4}$	0.203–0.266	2414×
held-out invariance ratio	0.0027–0.0121	0.983–0.994	282×
pairing residual	0.829–0.865	0.934–0.957	n/a

What “no physical labels” means. The search sees only latent states and the model’s own one-step transition. It never sees θ , $\dot{\theta}$ or the energy E , and nothing in Equations 2 and 3 refers to them. Ground truth enters afterward, when we score the recovered C against E to report $|\rho|_E$. One hyperparameter is an exception worth stating: the extraction dimension was chosen with reference to energy correlation, on a sweep over three models that are not among the three we report. It was then fixed and applied unchanged here, which is why Section 3 describes it as fixed before these models were evaluated.

Reference energy. We compare C with the textbook pendulum energy

$$E = \frac{1}{6}\dot{\theta}^2 + 5 \cos \theta.$$

Gymnasium uses a semi-implicit integrator, so it conserves a nearby shadow Hamiltonian $\tilde{H} = H + O(\Delta t)$ rather than E itself [Hairer et al., 2006]. In our trajectories, this discretization produces about 12% relative oscillation in E without secular drift. We treat that mismatch as a noise floor.

Earlier recurrent-model study. We developed the extraction method on a smaller recurrent model before applying it to DreamerV3. That work is unpublished, so it does not count as evidence for the DreamerV3 results. Appendix A summarizes it only to document the origin of the method and the controls that we retest here.

Code and run logs. The code, committed run logs used to generate every figure, and the paper source are available at <https://github.com/Zarand3r/latent-noether-paper1>. The figures regenerate from the committed logs without a GPU.

3 Training makes invariant recovery reproducible

At the extraction dimension fixed in advance, the three trained conservative models recover scalars with $|\rho|_E = 0.973, 0.967$, and 0.975 . Along observation-conditioned trajectories, the recovered C has within-trajectory variance equal to 1.1×10^{-4} of its total variance. It is therefore nearly constant within a trajectory while still varying across trajectories. Table 1 reports this quantity as the drift of C . The held-out invariance ratio in the next row evaluates the invariance-only candidate rather than the jointly fitted C , so the two statistics should not be compared as if they measured the same object.

The energy correlation alone is weak evidence [Hewitt and Liang, 2019, Belinkov, 2022]. Our search favors functions that remain constant within a trajectory and differ across trajectories. For a one-degree-of-freedom conservative pendulum, energy is essentially the only nontrivial scalar with that property. A latent representation that preserves enough information about $(\theta, \dot{\theta})$ can therefore produce a strong energy-correlated readout even if the model never learned to use energy dynamically.

The untrained baseline confirms this concern. Applying the identical pipeline to six randomly initialized DreamerV3 models gives $|\rho|_E$ values from 0.170 to 0.908. Four of the six exceed the 0.7 success threshold that we had inherited from the earlier architecture. That threshold therefore does not identify learning in DreamerV3.

The trained and untrained groups still differ in two ways. Every trained model scores above every untrained draw in energy correlation (0.967 versus 0.908 at the closest pair), and every trained model has

a lower pairing residual than every untrained draw (0.865 versus 0.912). With only three trained seeds and six random draws, these non-overlaps are suggestive rather than statistically decisive, so we do not attach a p -value to them.

The more robust difference is reproducibility. The three trained models cluster tightly between 0.967 and 0.975, whereas the six untrained models span 0.170–0.908. Training therefore appears to make the recovered scalar reproducible across independently trained models rather than merely making energy decodable.

4 The same search fails under matched damping

A search that produces a conserved-looking scalar for almost any latent representation could explain Section 3 without telling us anything about the learned dynamics. We test that possibility with matched models trained on a damped pendulum.

For the damped system, every trajectory converges to the same fixed point. Any continuous scalar that stays constant along all trajectories in the basin must therefore be constant throughout that basin, so the state has no nontrivial first integral for our search to recover.

Choosing the damping level. Very strong damping creates a separate problem: trajectories collapse into a small neighborhood of the fixed point, making the latent representation rank-deficient and creating directions that look constant for trivial reasons. We therefore fixed the damping level using only simulator states, before training any damped model. We chose the largest damping ratio whose late-window spread across trajectories remained above 25% of the conservative reference.

A stronger value, $\zeta = 0.15$, leaves 0% of that spread and fails this criterion. The rule selects $\zeta = 0.03$, which removes 17% of the energy per period while retaining 34% of the conservative spread. All three damped models pass the same training checks as the conservative models.

The conservative and damped models separate on all four statistics. Table 1 shows the result. The recovered C varies about $2400\times$ more within damped trajectories than within conservative ones. The damped models also have a held-out invariance ratio of 0.983–0.994, compared with 0.0027–0.0121 for the conservative models. Within the degree-4 family we search, the damped models contain no approximately conserved candidate.

The two arms use the same architecture, training budget, observation modality, and extraction procedure. The difference is the dynamics in the training data. This control does not prove that a damped model contains no conserved quantity of any possible form; it shows that our pipeline does not return an approximately conserved degree-4 scalar when the underlying dynamics remove the corresponding conservation law.

5 Correcting invariant drift improves rollouts

The recovered C stays nearly constant along observation-conditioned trajectories, but it drifts when the model rolls forward on its own. If that drift reflects a real model error, correcting it should improve prediction. If C is only a fitted statistic, enforcing it should simply perturb the rollout.

At every imagined step, we project the latent state back toward the level set defined by its initial value:

$$z \leftarrow z - \alpha(C(z) - C_0) \frac{\nabla C(z)}{\|\nabla C(z)\|^2}, \quad (4)$$

where C_0 is the value at the first encoded state. We map the correction back through P , leaving the part of h outside the extracted subspace unchanged. The intervention runs inside the imagination loop; it does not modify the decoder, and we never retrain the model.

Scoring. We use the same fixed grid $\alpha \in \{0, 0.05, 0.1, 0.2, 0.4\}$ for every arm and score the slope of rollout error across the full grid. Selecting the best α after seeing the curve would give each arm five chances to look good; the slope uses all five points. We measure rollout quality by pixel mean squared error against held-out video over 50 imagined steps.

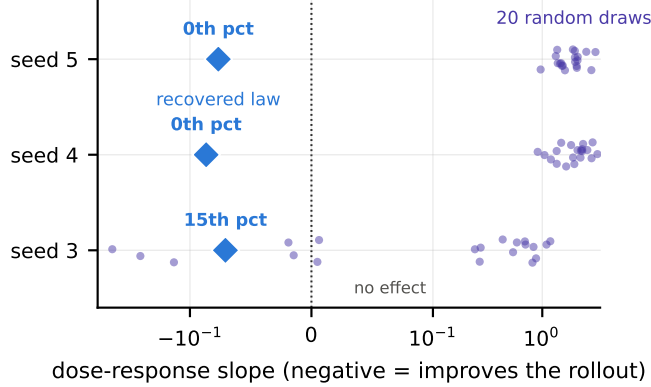


Figure 2: Each row compares one model with its own null distribution. The 20 small marks are norm-matched random degree-4 constraints, using the same draws across models; the diamond is that model’s recovered constraint. No random constraint beats the recovered one on seeds 4 and 5. On seed 3, seven random constraints reduce error and the recovered constraint ranks 15th of 20.

The scalar C is fitted on the same 52 held-out trajectories used to evaluate the edit, so the absolute intervention numbers are within-sample for the fitted scalar. The random-constraint arm uses the same evaluation setting, so the comparison between recovered and random constraints remains matched.

Controls. We compare three interventions. First, we enforce each conservative model’s own recovered C . Second, we enforce a random polynomial from the same degree-4 basis, matched in norm. Third, we enforce each damped model’s own recovered C . We draw 20 random polynomials and apply the same 20 to each of the three conservative models, giving 60 model–constraint evaluations from 20 distinct random constraints.

Figure 1c shows the dose response. At the strongest projection, enforcing the recovered constraint lowers rollout error by 2.9%, 3.3%, and 3.5% across the three conservative models. Across the 60 random-constraint evaluations, the median change is +53.9%, and only 7 of 60 lower the error. Applying the same kind of edit to the damped models changes error by only +0.5%. The recovered arm also behaves much more consistently across seeds: its per-model slopes span 0.016, compared with 3.71 across random evaluations.

The specificity result is not uniform. Figure 2 compares each model with its own 20-draw null. For two models, no random constraint performs better than the recovered one. For the third, 7 of 20 random constraints also lower error, and the best random draw reaches −21.9%, compared with −2.9% for the recovered constraint. We do not know why this seed behaves differently.

The intervention result is therefore narrow but consistent: enforcing the recovered invariant improves all three conservative models by 2.9–3.5%, while matched random constraints usually hurt. The improvement is specific to the recovered invariant on two models, but not on the third.

6 The correction acts on causally important directions

Section 5 shows that enforcing the recovered scalar can improve rollout accuracy. We next ask which latent directions the intervention changes.

For each direction u in the extracted subspace, we measure its variance and the damage caused by perturbing the latent state along it:

$$V(u) = \text{Var}(u^\top z), \quad D_H(u) = \mathbb{E}[L_H(z + \epsilon u) - L_H(z)], \quad (5)$$

where L_H is the model’s H -step imagination error against held-out video. V is the quantity PCA uses to rank directions. D_H measures causal sensitivity of the rollout and does not require simulator state. We use $\epsilon = 0.25|z|$, based on the calibration in Appendix B.

A perturbation injected into the latent state persists during imagination rather than decaying away. After 40 steps, the perturbation has grown to 6.3 times its initial size for $\epsilon = 0.05|z|$, 2.8 times for $0.25|z|$, and 1.9

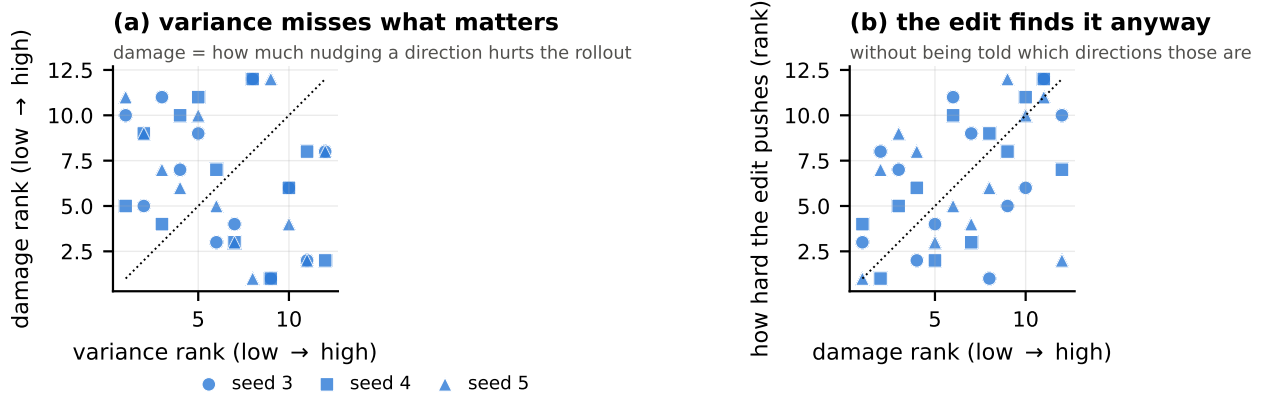


Figure 3: Each point is one of the 12 extracted directions, ranked within its own model; the dotted line marks equal ranks. **(a)** Variance rank versus damage rank at a 40-step horizon. **(b)** Damage rank versus the magnitude of the invariant correction along that direction.

times for $0.5|z|$, taking the median across the three models. Larger perturbations amplify less. Small latent errors can therefore produce substantial rollout differences over the horizons we study.

Variance and causal importance disagree at short horizons. At a 40-step horizon, variance rank and damage rank are anticorrelated on all three models: -0.364 , -0.259 , and -0.406 . The damage caused by the most and least consequential directions differs by 13.5–118.8 times across models (Figure 3a). The damage ranking itself is reproducible: splitting the evaluation trajectories in half gives a median rank correlation of 0.95 across the settings we tested, with a minimum of 0.52 at the smallest perturbation and longest horizon.

This disagreement weakens at longer horizons. Pooling three models and three perturbation sizes, the median $\rho(V, D_H)$ is -0.252 at horizon 20 and is negative in all 9 settings; -0.196 at horizon 50 and negative in 5 of 9 settings; and -0.028 at horizon 100 and negative in 5 of 9 settings, with one setting reaching $+0.727$. Once the rollout has already diverged substantially, high-amplitude directions can dominate total error and the rankings begin to realign.

We therefore make a limited claim: at short horizons, PCA variance can rank latent directions differently from their causal effect on prediction. We measure this only on the conservative models.

The invariant correction tends to push along high-damage directions. We also measure how strongly the projection in Equation 4 pushes along each latent direction. That quantity correlates with D_H at $+0.385$, $+0.692$, and $+0.287$ across the three models (Figure 3b). The projection was never given the damage ranking. The correlations remain well below 1, so the correction is only partly concentrated on the most consequential directions.

6.1 Basis-independent properties reproduce better than coordinate-level selectors

The three independently trained models agree closely on several quantities that do not depend on a particular latent basis. Their energy-recovery scores vary by only 0.008, their intervention slopes by 0.016, and the top-3 eigenvalue mass of

$$M_C = \mathbb{E} \left[\frac{gg^\top}{\|g\|^2} \right], \quad g = \nabla C,$$

by 0.046, with values 0.448, 0.441, and 0.487.

Selectors defined on individual latent coordinates are much less stable. Ranking coordinates by ∇C and keeping the top three gives intervention slopes of -0.075 , $+0.016$, and -0.089 : this beats a variance-based ranking on two models, loses on the third, and changes sign across seeds. Replacing the coordinate mask with an orthogonal projector onto the top eigenspace of M_C beats the variance selector on all three models.

On one model, however, it falls only at the 41st percentile of 100 random rank-3 subspaces. It is therefore better than variance on that model without being distinguishable from a random subspace of the same rank.

The spectrum of M_C also shows that three of the twelve directions contain just under half of its total mass. Because the correction has rank one at each state, this spectrum measures how much ∇C rotates as the state changes. The amount of rotation is similar across all three models.

These measurements are consistent with the three models representing the same scalar invariant in different latent coordinate systems. We treat that as an interpretation, not a direct measurement. Raw latent bases are arbitrary across independently trained models, so comparing coordinate-aligned subspaces would not be meaningful without an explicit alignment. What reproduces here are basis-independent properties of the scalar, not a fixed set of latent coordinates.

7 Ablations and limitations

The flow-generation term mainly reduces seed-to-seed variation. Our default extraction asks C both to remain constant along trajectories and to satisfy $F \approx B\nabla C$ with antisymmetric B . We isolate the second condition by holding the candidate family, data, dimension, polynomial degree, and α grid fixed and selecting C either by conservation alone or by the joint criterion.

Conservation alone recovers energy at 0.950, compared with 0.973 for the joint rule, and gives a slightly better average intervention slope (-0.087 versus -0.077 , winning on two of three models). The main difference is reproducibility: recovery varies by 0.035 across seeds under conservation alone and by 0.008 under the joint rule. Conservation drives both recovery and intervention performance; the flow-generation criterion contributes little to mean accuracy but reduces seed-to-seed variation.

The pairing residual is a diagnostic, not a recovery score. As the extraction dimension increases from 6 to 12, energy recovery improves from 0.721 to 0.967 while the pairing residual worsens from 0.738 to 0.876. Recovery already reaches 0.967 at dimension 8, where the residual is lower at 0.813.

The extra dimensions do not explain this pattern by carrying energy while resisting the Hamiltonian fit. Directions 7–12 account for 44.9% of the residual and 43.5% of the flow, nearly the same share. The pairing residual therefore does not track recovery quality across extraction dimension, so we use it only as a diagnostic. Appendix B reports the full sweeps.

Energy information can lie outside the highest-variance directions. On two of the three models, principal directions 7–12 carry about twice as much energy information as the top six directions: 2.23 times and 2.13 times as much. On the third model, the ratio reverses to 0.91. We report the three models separately because a median would hide that sign change. This result is consistent with the causal analysis in Section 6, but it is not uniform across seeds.

The frequency-weighted mechanism remains unresolved. Accumulated invariant error predicts which rollouts degrade most. An integral predictor reaches $R^2 = 0.187$, compared with 0.021 for a fitted power law in time at the horizon where both perform best, and shuffling destroys the relation.

The current sample cannot tell whether the relevant integral needs the frequency weighting found in the earlier recurrent model or whether an unweighted integral is enough. In a nested comparison with 400 bootstrap resamples, the weighted term adds explanatory power beyond the unweighted term in 1 of 9 seed-horizon combinations; the unweighted term adds beyond the weighted term in 0 of 9. At $n = 512$, we therefore treat the mechanism as unresolved. Appendix A describes the stronger result from the earlier recurrent model.

Scope. This study covers one physical system, one degree of freedom, one architecture, and a much shorter training schedule than standard DreamerV3. The pendulum trajectories cross the separatrix: 17% rotate rather than librate, so we restrict the mechanism analysis to the librating trajectories. We have not tested contacts, multiple interacting objects, nonzero actions, or systems whose frequency varies enough to identify the frequency-weighting mechanism. Sections 6 and 6.1 use only the conservative models.

8 Related work

Probing. Linear and nonlinear probes can decode physical or linguistic variables from hidden states [Alain and Bengio, 2017], but probe accuracy mixes information stored by the representation with the capacity of the probe itself. Control tasks [Hewitt and Liang, 2019] and later critiques [Belinkov, 2022] make this limitation explicit. Our untrained DreamerV3 baseline plays the same role for invariant extraction: four of six randomly initialized models exceed the success threshold we had previously used. Work on emergent world representations in sequence models [Li et al., 2023, Nanda et al., 2023] goes beyond decoding by intervening on learned representations. Our rollout intervention in Section 5 follows that logic.

Equation discovery and structured dynamics models. Equation-discovery methods such as SINDy [Brunton et al., 2016] infer governing equations directly from observed trajectories. Hamiltonian and Lagrangian neural networks [Greydanus et al., 2019, Cranmer et al., 2020] instead build physical structure into the architecture so that conservation holds by construction. These approaches model the physical system itself. We treat an already trained world model as the object of study and ask what its learned transition approximately preserves. Our extraction also uses standard linear-algebraic ideas related to Koopman and dynamic mode methods [Schmid, 2010].

Evaluation of world models. Dreamer-style world models [Hafner et al., 2023, Ha and Schmidhuber, 2018] and planning-based learned models [Schrittwieser et al., 2020] are usually evaluated by prediction quality or downstream return. Those metrics do not show whether the latent transition preserves physical invariants. The disentanglement literature [Locatello et al., 2019] similarly shows how apparent structure in a representation can survive weak controls. Our damped models directly test whether the invariant-search procedure still returns a conserved-looking scalar when the training dynamics have no nontrivial first integral of the kind we seek.

The dynamical-systems tools we use are standard. The main result is empirical: a pixel-trained DreamerV3 develops a reproducible energy-like invariant with measurable dynamical relevance, and matched controls rule out simple decodability as the explanation.

9 Conclusion

A DreamerV3 trained only on pendulum video develops a reproducible energy-like invariant. Energy correlation alone does not establish that result: randomly initialized models can also produce high correlations. The stronger evidence comes from three independent tests. The recovered scalar is approximately conserved along observation-conditioned trajectories in conservative models; the same search finds no approximately conserved degree-4 scalar in matched damped models; and correcting the scalar’s drift during autonomous imagination improves prediction. The intervention lowers rollout error by 2.9–3.5% on all three trained conservative models, although it beats matched random constraints on only two of the three.

The result also exposes a gap between representation and transition. The model learns a scalar that behaves like an invariant on real trajectories but does not preserve it perfectly when it imagines forward. A natural next question is whether world-model transitions can preserve such learned structure by construction, and whether the same effect survives in systems with actions, contacts, and multiple interacting objects.

A Earlier recurrent-model study

The extraction method was developed on a smaller recurrent model before we applied it to DreamerV3. This study is unpublished and does not support the DreamerV3 claim directly. We include it to document which findings motivated the controls in the main paper and which findings failed to transfer.

A.1 Setup and extraction

We trained a single-layer gated recurrent unit with 48–64 hidden units, about 13k parameters, to predict the next observation in low-dimensional mechanical systems. The model received position but not velocity, so its

recurrent state had to infer velocity from the sequence. We studied two conservative systems, an anharmonic central-force system and a pair of coupled anharmonic oscillators, and matched dissipative variants. The extraction procedure never used physical labels.

Let z denote a low-dimensional representation of the hidden state and f the latent flow induced by the trained transition. We jointly searched for a scalar C and an antisymmetric field B satisfying $f \approx B\nabla C$. Because B is antisymmetric, $\nabla C^\top B \nabla C = 0$, so a good fit implies that C changes little along the flow. We score the fit with the normalized residual

$$R = \frac{\|f - B\nabla C\|^2}{\|f\|^2}.$$

A.2 Results

A randomly initialized recurrent network already contained a latent scalar with $|\rho| \approx 0.998$ correlation with true energy. A high-dimensional random representation of a low-dimensional trajectory can preserve enough state information for a flexible readout to reconstruct energy. This result motivated the untrained control in the DreamerV3 study.

With the joint criterion, the recovered scalar reached $|\rho_E| \geq 0.972$ on conservative models, sometimes approaching 0.999. The residual separated conservative and dissipative models in a representative comparison ($p = 5.4 \times 10^{-6}$, $d = 3.10$). During free rollout, correcting the hidden state to keep C near its initial value reduced trajectory error by up to 68% and true-energy drift by 69%; the same intervention increased error on dissipative models.

About 77% of rollout error lay along the true physical flow, while only 14% changed the conserved quantity. The model therefore tended to stay near the correct orbit while moving along it at the wrong rate. For a nonlinear oscillator, frequency depends on the conserved quantity, so a small energy error can create a persistent frequency error that accumulates into phase error. We tested the predictor

$$X(t) = \int_0^t \kappa(E_s) \Delta E(s) ds, \quad \kappa(E) = \frac{d \ln \omega}{dE},$$

against generic functions of time on held-out trajectories.

predictor	central force	coupled oscillators
$a + bt$	0.124	-0.028
$a + bt^p$ (best p)	0.131	-0.024
$a + b \int \kappa \Delta E ds$	0.658	0.358
plus an oscillatory correction	0.703	—
plus an angular-momentum term	0.657	—
fitted coefficient b	0.91	0.67

The fitted coefficient has roughly the scale predicted by the mechanism, and κ comes from the simulator rather than from a fit. Adding angular momentum does not improve the central-force predictor, and the coupled system has no rotational symmetry, so a scalar-energy account is sufficient for these errors. A one-time intervention on the invariant also changed later phase drift with the predicted sign in 94% of cases, with held-out $R^2 = 0.887$.

A.3 Negative results

The invariant gradient is not the physical energy gradient. Matching perturbations so that each direction produces the same instantaneous change in true energy does not make their downstream effects equal. Perturbing along ∇C changes later phase drift the most per unit energy change, a random direction less, and a direction tangent to C the least, on every seed. The size of the gap depends on how the coefficient is estimated, so we report only the ordering. Separately, ∇C changes true energy only 1.08 times more per unit displacement than a random direction. Recovering the right scalar does not imply that the surrounding latent geometry matches the physical state space.

The learned model breaks a physical symmetry. For a rotationally symmetric central-force system, radial frequency must be even in angular momentum: $\omega_r(E, L) = \omega_r(E, -L)$. At fixed energy, the simulator obeys this constraint. $|L|$ alone predicts radial frequency with $R^2 = 0.89$, while signed L adds no explanatory power ($R^2 = 0.00$). The learned model violates the symmetry: signed L predicts its radial frequency with $R^2 = 0.30$. Recovering an energy-like scalar therefore does not imply that the model has recovered the full governing law.

Structural explanations that did not hold. A low-dimensional part of the hidden state carried most of the information used to decode future observations, which suggested a quotient interpretation. The corresponding divergence statistic, however, separated conservative from dissipative systems at only AUC = 0.68, so we treat the observation as descriptive rather than explanatory. A subspace selected by transition-closure criteria also failed to outperform PCA at matched dimension. Finally, extraction depends on the representation: whitening can destroy recovery, and restricting the analysis to only the highest-variance directions can remove physical information.

A.4 Comparison with DreamerV3

finding	earlier model	DreamerV3 (this paper)
energy-like invariant recoverable	$ \rho_E \geq 0.972$	replicates, ≈ 0.97
untrained networks score high	up to 0.998	replicates, up to 0.908
conservative and dissipative models separate	$p = 5.4 \times 10^{-6}$	replicates, no seed overlap
enforcing the invariant improves rollouts	up to 68%	same direction, 2.9–3.5%
residual indicates recovery quality	holds	does not hold
frequency weighting improves the fit	$R^2 = 0.658$ vs 0.131	unresolved at $n = 512$
∇C is the physical energy direction	false	remains unsupported
a correct scalar implies correct geometry	false	remains unsupported

The intervention transfers in direction but not in magnitude, and the percentages are not directly comparable. The earlier study measures trajectory error in a two-dimensional state space, while this paper measures pixel error against video. Much of each video frame is static and can remain correct even when the latent dynamics drift, so the pixel metric naturally compresses the apparent effect size. Any remaining difference could come from the architecture, system, or extraction dimension.

The flow-generation residual also fails to transfer as a recovery metric. In the earlier recurrent model it separated the relevant cases; in DreamerV3 it sits near the middle of the random-polynomial distribution, while conservation provides the stronger signal.

B Reproducibility notes

Training details. We set free bits to 0 rather than DreamerV3’s reference default of 1.0. The measured KL divergence settles near 1.8 nats, well above collapse, so the floor is inactive in this setting. We report the change because otherwise describing the setup as the published defaults would be inaccurate. All three trained models pass the acceptance checks; none were discarded.

Extraction data. Training uses trajectories 0–203 from the 256-trajectory dataset. Extraction uses the remaining 52 trajectories. We fit C on those 52 held-out trajectories and evaluate the Section 5 intervention on the same 52. The recovered and random-constraint arms use the same data split.

Calibrating the displacement. The sensitivity analysis in Section 6 needs a perturbation large enough to produce a measurable nonlinear effect. At $\epsilon = 0.05|z|$, damage is about -0.5% of baseline and its sign varies across directions. Because imagination is deterministic, repeated unperturbed rollouts are identical, so these small values do not come from sampling noise.

For the leading direction, damage changes by -0.5% , $+3.1\%$, $+18.1\%$, $+46.4\%$, and $+128.8\%$ at $\epsilon/|z| = 0.05, 0.10, 0.25, 0.50$, and 1.00 . We use $\epsilon = 0.25|z|$, the smallest perturbation whose damage exceeds 10% of baseline while the response still grows smoothly.

Scoring the intervention over a grid. We score the slope of rollout error over the fixed α grid rather than the minimum value on that grid. Taking the minimum would select the most favorable of five points after observing the curve and can misrepresent the overall response. In one diagnostic run, that rule would have reported a 5.9% improvement even though error rose monotonically to twice baseline across the sweep. The slope avoids that failure mode.

Testing frequency weighting. The weighted and unweighted integral predictors are nested, so comparing their standalone R^2 values is not enough. A difference of 0.019 in R^2 , for example, does not establish which predictor is better unless we compare that increment with its sampling uncertainty. The weighted predictor leads in 6 of 9 seed-horizon combinations, but the nested bootstrap test in Section 7 finds a significant incremental contribution in only 1 of 9. We therefore report the frequency weighting as unresolved rather than supported or refuted.

Extraction dimension and residual decomposition. Figure 4 reports recovery and pairing residual across extraction dimension. Figure 5 shows where energy information, latent flow, and pairing residual lie across the high- and lower-variance principal directions.

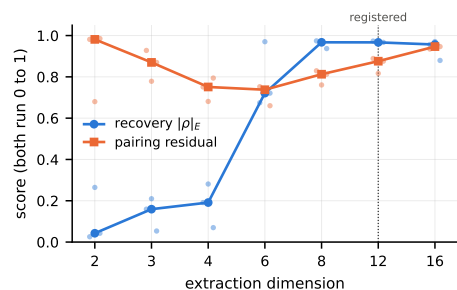


Figure 4: Energy recovery and pairing residual versus extraction dimension. Lines show medians over the three models and points show individual models. Both quantities are dimensionless and plotted on the same 0–1 axis.

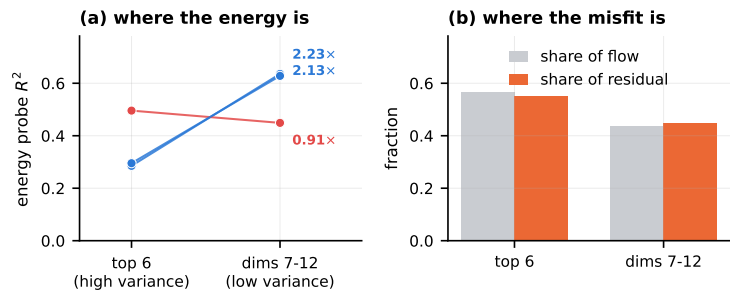


Figure 5: (a) Energy-probe R^2 for the top six principal directions versus directions 7–12, shown separately for each model. (b) Fraction of latent flow and pairing residual carried by each group, using medians across the three models.

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